Helmet and Reflective Vest Detection for Construction Safety Compliance

PPE Compliance Monitor — A real-time computer vision-based safety monitoring system developed for deployment across construction and industrial sites in the **Ambur–Vellore** region.

COMPUTER VISION | YOLOV8 | CONSTRUCTION SAFETY

Problem Statement

Across construction and industrial sites, workers frequently operate without mandatory Personal Protective Equipment (PPE) such as helmets and reflective vests. This widespread non-compliance poses significant risks to worker safety and site management alike.

Challenges in Manual Monitoring

Traditional safety supervision relies entirely on human inspectors, which introduces critical weaknesses in the compliance process.

- Inconsistent enforcement across shifts and zones
- Time-consuming manual walkthroughs on large sites
- Not scalable to cover multiple camera feeds or large worksites simultaneously
- Prone to human error, fatigue, and oversight

Consequences of Non-Compliance

The absence of automated monitoring mechanisms leads to measurable harm at both the individual and organisational level.

- Elevated probability of workplace injuries and fatalities
- Poor compliance tracking with no audit trail
- Regulatory penalties and legal liability for site owners
- Reactive rather than preventive safety management

The need for an automated, scalable, and accurate PPE detection system is therefore both urgent and well-justified.

Objective

This project aims to design and deploy an intelligent, vision-based system that addresses the shortcomings of manual PPE monitoring. The following core objectives guide the development of the system:

1. **Automated PPE Detection**
   Detect the presence or absence of helmets (hardhats) and reflective safety vests on workers in real time using deep learning-based object detection.

2. **Violation Flagging**
   Identify and flag instances of PPE non-compliance immediately, enabling prompt corrective action before accidents occur.

3. **Continuous Monitoring**
   Provide uninterrupted surveillance through camera systems across the site, eliminating dependence on periodic manual inspections.

4. **Safety Analytics**
   Generate measurable and trackable safety metrics — such as compliance scores, violation frequency, and alert logs — to support data-driven decision-making.

5. **Reduced Human Dependency**
   Minimise reliance on human supervisors for routine monitoring, allowing safety personnel to focus on critical interventions.

Literature Survey

A review of existing research reveals that PPE detection has been an active area of study within computer vision. Deep learning approaches, particularly Convolutional Neural Networks (CNNs) and YOLO-family models, have demonstrated strong results in identifying safety equipment on construction sites.

State of Existing Research

YOLO-based approaches are widely preferred for real-time detection tasks due to their speed and accuracy trade-off. Most systems focus on detecting helmets, with some extending to vests and other PPE classes. These works confirm the feasibility of vision-based safety monitoring at construction sites.

Identified Limitations

- **High false positive rate:** Many systems incorrectly classify background objects as PPE
- **Frame-by-frame flickering:** Detections appear and disappear inconsistently across video frames
- **Limited detection classes:** Most systems detect only one PPE type (helmet *or* vest), not both simultaneously
- **No integrated dashboard:** Results are rarely tied into a monitoring or analytics interface

Proposed Improvements

This project directly addresses the gaps identified in the literature through the following enhancements:

Multi-Class Detection

Simultaneous detection of helmets, vests, persons, and machinery within a single inference pass

Temporal Buffering

Stabilises predictions across multiple consecutive frames, eliminating detection flickering

Integrated Analytics

A real-time dashboard with alert logging, compliance scoring, and incident reporting

Proposed System

The PPE Compliance Monitor is an AI-based system designed for seamless integration into existing construction site infrastructure. It accepts multiple input modalities and delivers actionable, annotated output through a real-time interface.

1

Input Sources

- Live camera feed (CCTV / IP camera)
- Image upload for static analysis
- Video upload for recorded footage review

2

Processing Engine

- Object detection via YOLOv8
- Confidence threshold filtering to suppress weak detections
- Non-Maximum Suppression (NMS) for clean bounding boxes
- Temporal frame buffering for stability

3

Output Delivery

- Bounding box annotations on detected persons and PPE
- Violation highlights for **NO-Hardhat** and **NO-Vest** classes
- Real-time alerts and timestamped logs
- Dashboard metrics and compliance score

Methodology

Dataset Details

The model was trained on a comprehensive construction site dataset sourced from publicly available and curated repositories. The dataset encompasses diverse lighting conditions, worker postures, and PPE states.

2,801

Total Images

Covering varied site conditions

38,352

Annotations

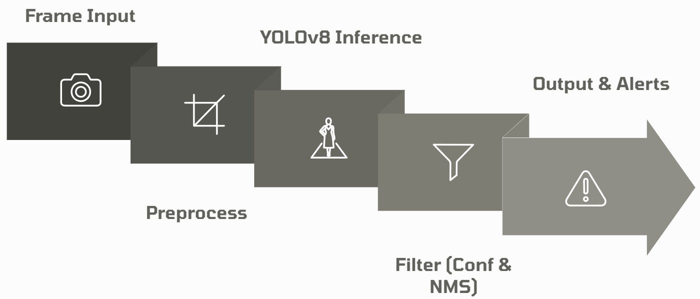
Bounding boxes across all classes

Training Configuration

Parameter	Value
Model	YOLOv8m (medium)
Epochs	100
Image Size	640 × 640 px
Batch Size	32
Optimizer	SGD, momentum 0.937

Processing Pipeline

Each frame or uploaded input passes through a structured multi-stage pipeline before results are presented to the user.



The pipeline ensures that only high-confidence, temporally stable detections are surfaced to the monitoring dashboard, significantly reducing false alarms.

System Architecture

The system is built on a modular, full-stack architecture that separates concerns across the frontend, backend, detection engine, and data storage layers — ensuring scalability, maintainability, and ease of deployment.

Frontend

React + TypeScript — Provides an interactive dashboard for live visualisation, compliance metrics, and alert management. Designed for clarity and ease of use by site supervisors.

Backend

FastAPI — Handles all API requests, manages input routing (camera / image / video), and orchestrates communication between the frontend and the detection engine.

Detection Engine

YOLOv8 Model — Core inference engine responsible for real-time object detection, class prediction, and bounding box generation across all input types.

Data Storage

JSON-Based Storage — Persists incident records, safety metrics, and session state in a lightweight format suitable for the deployment environment without requiring a full database server.

Data Flow: User Input → Backend API (FastAPI) → YOLOv8 Detection Engine → Detection Results → Frontend Dashboard (React)

Key Contributions

This project makes several meaningful technical and applied contributions to the domain of automated safety monitoring. Each contribution addresses a specific gap or limitation identified during the literature survey and problem analysis.

1 End-to-End PPE Compliance System

Developed a complete, deployable system covering the full pipeline — from raw camera input to annotated output and logged incidents — with no manual intervention required during operation.

2 Multi-Modal Real-Time Processing

Integrated support for live camera feeds, image uploads, and video uploads, making the system versatile across different site monitoring scenarios.

3 Temporal Buffering for Stability

Implemented a multi-frame consistency mechanism that tracks detections across consecutive frames, reducing false alarms and eliminating flickering common in single-frame inference systems.

4 Configurable Detection Thresholds

Enabled site-specific tuning of confidence and Intersection over Union (IoU) thresholds, allowing operators to balance sensitivity and precision based on their deployment environment.

5 Analytics Dashboard and Incident Logging

Designed a dedicated monitoring dashboard providing compliance scores, alert frequency, and violation timelines, alongside persistent incident logs with timestamps for audit and review purposes.

Results and Key Metrics

The trained YOLOv8m model was evaluated on a held-out test set. The results demonstrate strong detection performance across all PPE classes, with inference speeds suitable for real-time deployment.

Model Performance

0.88

mAP@0.5

Mean Average Precision at IoU threshold 0.5

0.92

Precision

Proportion of correct positive detections

0.81

Recall

Proportion of actual violations detected

23.5ms

Inference Speed

Per image on standard hardware

Operational Output Metrics

Beyond model accuracy, the system tracks several operational indicators that reflect real-world safety performance on site:

- **Detection Accuracy:** Percentage of PPE items correctly identified per session
- **False Alarm Rate:** Frequency of incorrect violation flags, minimised via temporal buffering
- **Alerts per Hour:** Volume of compliance violations surfaced per monitoring hour
- **Safety Compliance Score (%):** Aggregate metric reflecting overall PPE adherence across the monitored workforce

Output Artefacts

- Annotated images and video frames with bounding boxes and class labels
- Timestamped violation records persisted in the incident log
- Real-time dashboard updates reflecting live compliance status

Conclusion and Applications

Conclusion

The PPE Compliance Monitor successfully demonstrates that automated, vision-based safety monitoring is both technically feasible and practically effective. By leveraging YOLOv8m with temporal stabilisation and an integrated analytics dashboard, the system provides accurate, real-time detection of helmet and reflective vest compliance with minimal human intervention.

- Achieves strong model accuracy (mAP@0.5: 0.88, Precision: 0.92)
- Eliminates inconsistencies associated with manual monitoring
- Improves compliance tracking with persistent, auditable records
- Scalable to large multi-camera construction environments

Applications and Impact

Construction Sites

Primary deployment in the Ambur–Vellore corridor; adaptable to any active construction zone

Industrial Workplaces

Factories, warehouses, and heavy industry facilities requiring mandatory PPE compliance

Smart Surveillance

Integration with existing CCTV infrastructure for smart city and campus safety applications

Outcome

- Increased worker safety through proactive violation detection
- Reduced risk of fatal accidents and regulatory non-compliance
- Efficient, scalable, and data-driven compliance monitoring